

STEM Fusion

Introduction to Quantum Computing

The Core Ideas

Classical vs Quantum · Qubits · Superposition · Entanglement

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Innovation Hall 129

Schedule



Homeroom

10:00 – 10:15



G1 – Session Block

10:15 – 11:35



Lunch

11:40 – 12:35



G2 – Session Block

12:40 – 2:00

Student Expectations



NO FOOD



NO DRINKS



Be Curious



Respect Others



Ask Questions



Try & Fail



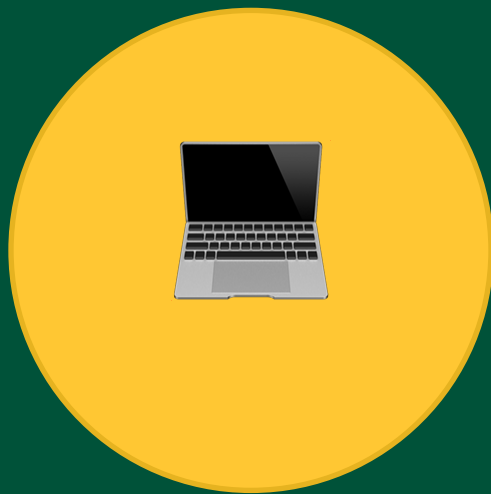
No Phones



Stay Focused



Have Fun!



Classical vs Quantum

Not just faster — a completely different kind of machine

How Classical Computers Work

Every classical computer — from your phone to a supercomputer — stores ALL information as 1s and 0s called bits.



A bit is a light switch

It can only be ON (1) or OFF (0). Never anything in between. Your phone has billions of tiny switches called transistors flipping billions of times per second to make every app, photo, and video work.



Classical computers work step by step

One operation at a time, in sequence. The same input always gives the same output. There is no randomness or ambiguity — everything is completely predictable.



Where they hit a wall

For some problems, the number of combinations to check grows so explosively fast that no classical computer — however powerful — could ever finish. It is not a speed problem. It is a fundamentally different kind of problem.

Problems Classical Computers Can Never Solve

Some problems grow so exponentially hard that adding more processors or waiting longer does not help. You need a completely different approach.



Breaking Codes

The encryption protecting every bank, website, and private message relies on a simple fact: multiplying two huge numbers together is easy, but working backwards to find those numbers is impossibly hard for classical computers.

A 2048-digit encryption key would take the world's fastest supercomputer longer than the age of the universe to crack.

→ A quantum computer could do it in hours.



Finding the Best Route

Finding the shortest route connecting 30 cities, visiting each once, involves more possible routes than there are atoms in the observable universe.

Classical computers can only check routes one by one — or make educated guesses.

→ Quantum computers can explore many possibilities at the same time.



Simulating Molecules

Designing medicine means understanding how molecules interact at an atomic level. But molecules follow quantum rules — simulating them classically is like modelling an ocean with a spreadsheet.

→ Quantum computers simulate quantum physics natively, because they ARE quantum systems themselves.

Classical vs Quantum — Side by Side

	Classical Computer	Quantum Computer
 Basic unit	Bit — only 0 OR 1	Qubit — 0, 1, or BOTH at once
 How it thinks	One possibility at a time	Many possibilities simultaneously
 Results	Exact and predictable every time	Probabilistic — needs many runs
 Temperature	Room temperature	Colder than outer space (-273°C)
 Best at	Email, games, apps, everyday tasks	Encryption, drug discovery, optimisation
 Maturity	Decades of refinement	Early commercial stage

*Quantum computers do not make classical computers obsolete
— they solve the problems classical computers are fundamentally unable to solve.*



Qubits

The quantum bit — spinning coins and impossible states

What is a Qubit?

A qubit is the quantum version of a bit.

Unlike a bit (always 0 or 1), a qubit can be 0, 1, or a mixture of both — all at the same time.



A regular bit is a coin lying flat

It is either heads (0) or tails (1). You know exactly what it is by looking. One definite value with no uncertainty.



A qubit is a coin spinning in the air

While spinning, it is neither heads nor tails — it is in superposition of both at once. It has a probability of being 0 and a probability of being 1. The moment it lands (is measured), it collapses to one or the other.



Qubits are made from real physical things

Scientists build qubits from individual electrons, atoms, or tiny superconducting loops. These objects genuinely behave quantum mechanically — confirmed in thousands of experiments worldwide.



Qubits need extreme cold to survive

Most quantum computers must be cooled to near absolute zero (-273°C) — colder than outer space. At room temperature, the delicate quantum behaviour collapses almost instantly.

Why Qubits Are Powerful — The Doubling Effect

*Each qubit you add DOUBLES the number of states the quantum computer can represent simultaneously.
The growth is explosive.*

Classical bits — one state at a time

1 bit 2 possibilities — but only 1 at a time

2 bits 4 possibilities — but only 1 at a time

10 bits 1,024 possibilities — but only 1 at a time

50 bits 1 quadrillion+ — but still only 1 at a time

Qubits — ALL states simultaneously

1 qubit 2 states simultaneously

2 qubits 4 states simultaneously

10 qubits 1,024 states simultaneously

50 qubits Over 1 QUADRILLION states simultaneously!

50 classical bits = ONE state. 50 qubits = 1,125,899,906,842,624 states — all at once.



Superposition

Being in two states at once — until you look

Superposition — Four Ways to Understand It

*Superposition means a qubit exists in multiple states at the same time
— not flickering between them, but genuinely in both simultaneously.*



The Spinning Coin

While a coin spins in the air it is not heads or tails — it is in superposition of both. The moment it lands (is measured), it becomes one or the other. The spinning is the superposition; landing is the measurement.



Radio Waves

Before you tune in, your antenna receives ALL stations at once — a superposition of every frequency. Tuning collapses it to one station. Measuring a qubit is like tuning the dial — you pick one outcome and the rest disappear.



Schrödinger's Cat

Physicist Erwin Schrödinger described a cat sealed in a box with a quantum trigger. Until you open the box, the cat is mathematically both alive AND dead — in superposition.



Why This Powers Computing

If a qubit is in superposition of 0 AND 1, one operation acts on BOTH values simultaneously. With many qubits in superposition, one operation explores an enormous number of possibilities at once. This is quantum computing's engine.

Measurement — When Quantum Becomes Classical

Before Measurement

The qubit exists in superposition — a combination of 0 and 1 simultaneously.

If it is an equal superposition:

- 50% chance of being 0
- 50% chance of being 1

Importantly — it is NOT secretly 0 or 1 and we just do not know which. It genuinely is both states at once, described by quantum probability waves.

This is where computation happens — quantum operations manipulate the superposition, changing the probabilities, before you ever look.

After Measurement

The qubit collapses — it becomes a definite 0 or 1, with the probability set by the superposition.

This is irreversible — the superposition is permanently gone.

⚠ Key consequence:

You only get ONE answer per run. To get reliable results, run the circuit thousands of times and see which answer appears most often.

This is why quantum algorithm design is hard — you must engineer the computation so the correct answer is the most probable result at measurement time.



Entanglement

Spooky action at a distance — Einstein's greatest puzzle

What is Entanglement?

Entanglement links two qubits so that measuring one instantly determines the other — regardless of the distance between them.

The Glove Analogy

Imagine putting one glove in each of two boxes and shipping them to opposite sides of the world. Opening yours and seeing 'left glove' instantly tells you the other is right.

That is classical correlation — determined from the start.

Entanglement is stranger: the gloves do not HAVE a handedness until you look. The moment you open yours, both gloves simultaneously become defined — anywhere in the universe, instantly.

Why Einstein Hated It

Einstein called it 'spooky action at a distance.' He believed particles must carry hidden information determining their states in advance — like gloves that already had a handedness.

But experiments have proven Einstein wrong. The particles genuinely do not have defined states until measured. The correlation is created AT the moment of measurement — instantly.

This has been confirmed repeatedly in labs worldwide. The universe IS that spooky.

NOT Faster-Than-Light

Entanglement does NOT let you send information faster than light.

Why? The result you get when measuring is RANDOM. You cannot control whether you get 0 or 1. So you cannot encode a message.

The correlation only shows up when both people compare their results — which requires normal (slow) communication.

Correlation does NOT equal communication.

Why Entanglement Matters for Computing

Quantum Parallelism

When qubits are entangled, they multiply rather than add.

2 entangled qubits → 4 states
10 entangled qubits → 1,024 states
50 entangled qubits → 1 quadrillion+

Entanglement creates complex superpositions that encode exponentially more information than separate qubits.

This is the engine behind quantum computing's power for certain problem types.

Unbreakable Encryption

Quantum Key Distribution (QKD) uses entanglement to share secret keys.

If a spy tries to intercept a quantum message, they must measure the qubits — and measuring changes them. The interference is immediately detectable.

This gives security guaranteed by the laws of physics, not just mathematical difficulty.

China already operates a 2,000 km quantum communication network using this principle today.

Quantum Teleportation

Yes — real, and experimentally demonstrated.

Scientists teleported the quantum state of a particle from Earth to a satellite 1,400 km away (China, 2017).

Important: it teleports quantum **INFORMATION**, not matter. No person or physical object moves — just the complete quantum description of a particle's state.

This is the foundation for a future quantum internet.